

Molecular Docking and ADMET Profiling of Cyanobacterial Natural Products Against the BRCA2 DNA-Binding Domain

A dual-receptor molecular docking and ADMET profiling pipeline
Target: Human BRCA2 DNA-binding domain (DSS1 binding site)
Ligands: Scytonemin, Phycocyanobilin, Lyngbyabellin A, Nostocarboline
Controls: Olaparib (affinity reference), Caffeine (negative)

This document outlines the complete computational workflow: receptor preparation, homology modelling, binding pocket detection, molecular docking across two receptor structures, interaction analysis, and ADMET profiling. It accompanies the data files in this repository.

1. Project Overview

This study computationally screens four cyanobacterial natural products — Lyngbyabellin A, Nostocarboline, Phycocyanobilin, and Scytonemin — for binding to the DNA-binding domain (DBD) of BRCA2, a tumour-suppressor protein central to homologous-recombination DNA repair. The compounds derive from cyanobacterial genera (*Nostoc*, *Anabaena*, *Lyngbya*) and were docked against the DSS1-binding site of BRCA2. Because no experimental crystal structure of the human BRCA2 DBD exists, the screening uses a dual-receptor strategy: an experimental mouse ortholog (PDB 1MJE) and a human homology model built from that same template via SWISS-MODEL. Results are compared across both tracks as a cross-species consistency check.

1.1 Tool Stack

Step	Tool	Type
Structure download	RCSB PDB	Web
Human receptor build	SWISS-MODEL	Web
Structure cleaning & visualization	PyMOL	Desktop
Binding pocket detection	PrankWeb / P2Rank	Web
Molecular docking	AutoDock Vina (via Tamarind Bio)	Web
Interaction diagrams	PLIP	Web
Drug-likeness & pharmacokinetics	SwissADME	Web
Toxicity profiling	ADMETlab 3.0	Web

2. Receptor Preparation

2.1 Mouse Receptor (Track 1) — PDB 1MJE

The crystal structure 1MJE (X-ray, 3.5 Å) was downloaded from RCSB PDB. It contains three chains: chain A (BRCA2 DBD, mouse, 600 resolved residues), chain B (DSS1, the co-crystallized peptide partner), and chain C (a 6-nucleotide oligo-dT ssDNA). In PyMOL, waters, DSS1 (chain B), heteroatoms, and the ssDNA (chain C) were removed. The ssDNA required explicit chain removal because it is recorded as ATOM rather than HETATM lines. The cleaned single-chain receptor was saved as 1mje_clean.pdb. Forty-nine residues are unresolved in the crystal structure and were noted as a limitation.

2.2 Human Receptor (Track 2) — SWISS-MODEL Homology Model

The human BRCA2 sequence (UniProt P51587 with 3,418 residues) was submitted to SWISS-MODEL. The 1MJE template (entry 1mje.1.B) was selected explicitly at 76.4% sequence identity. The resulting model covers residues 2478-3191 (714 residues) of the DBD.

Quality metric	Value	Interpretation
GMQE	0.03	Low because scored over the full 3,418-residue submission (18% coverage) — reflects coverage, not model quality
QMEANDisCo Global	0.64 ± 0.05	Primary quality indicator; scored for modelled region only. Moderate and trustworthy for a 76% identity, 3.5 Å template
Template identity	76.40%	1mje.1.B
Modelled range	2478-3191	714 residues, chain B

The template's ssDNA ligand was deliberately excluded by SWISS-MODEL ("binding site not conserved"), leaving a clean receptor pocket. The model was saved as BRCA2_SM.pdb.

2.3 Structural Alignment

The human model was superimposed onto the mouse receptor in PyMOL using the align command, yielding an RMSD of 0.324 Å over 3,913 atoms after iterative outlier rejection. This confirms both structures share a common coordinate frame, allowing a single grid box to target the equivalent pocket in both. Because the human model is templated on 1MJE, agreement between tracks is partly expected by construction — this is a cross-species consistency check rather than fully independent validation.

2.4 Hydrogen Removal for the Human Receptor

Tamarind Bio's internal PDB2PQR pipeline fails if the input PDB already contains hydrogen atoms (it misinterprets them as structural gaps and aborts). All hydrogens were therefore stripped from the aligned human model in PyMOL before upload, producing BRCA2_clean.pdb.

Note: The mouse receptor contained no hydrogens and required no such step.

3. Ligand Preparation

Six compounds were retrieved from PubChem as SDF files (accepted natively by Tamarind Bio; no MOL2 conversion required). Four are cyanobacterial test compounds; two are controls. Their SMILES were also recorded for the ADMET profiling step (SwissADME and ADMETlab 3.0 both require SMILES for running diagnosis).

[see "Molecular_Docking/03_ligands/SMILES_compounds.pdf"]

Compound	PubChem CID	Role	Source genus	MW (g/mol)
Lyngbyabellin A	10032587	Test ligand	Lyngbya	691.69
Nostocarboline	5326150	Test ligand	Nostoc	217.67
Phycocyanobilin	5280816	Test ligand	Anabaena / Nostoc	586.68
Scytonemin	135473381	Test ligand	Nostoc / Lyngbya	544.55
Olaparib	23725625	Affinity reference*	Synthetic (PARP inhibitor)	—
Caffeine	2519	Negative control	—	—

Note: Nostocarboline carries a formal positive charge (quaternary-nitrogen beta-carboline), which influences how Vina scores interactions with negatively charged residues (Asp, Glu).

***Olaparib control note:** Olaparib is a PARP1/2 inhibitor — it does not bind to BRCA2. It works by blocking PARP-mediated single-strand break repair, which in BRCA2-deficient cells causes lethal double-strand break accumulation (synthetic lethality). It has no known affinity for the BRCA2 DBD and was included here as a comparative affinity benchmark — a clinically approved, well-characterised small molecule docked against the same site — rather than a validated site binder. A true positive control would require a compound with experimentally confirmed BRCA2 DBD binding (e.g., a DSS1 mimetic), which is not available as a small-molecule SDF.

4. Binding Pocket Detection

PrankWeb (P2Rank) was used for pocket detection on 1mje_clean.pdb (chain A, default prediction model). The top-ranked pocket P1 was validated as the DSS1 binding site by measuring the distance from each pocket centre to the geometric centre of DSS1's original position in the raw structure: P1 sat 7.96 Å away (correct site) versus P2 which was 46.19 Å away.

Property	P1 pocket value
PrankWeb score	16.87
Probability	0.790
Number of residues	23
Distance to DSS1 centre	7.96 Å (confirms DSS1 site)
Pocket centre (X, Y, Z)	27.93, 155.71, 62.56

Because both receptors share the same coordinate frame (Section 2.3), the same pocket and grid box apply to both tracks; no separate detection was needed for the human model.

4.1 Grid Box Parameters

Centre X = 27.93, Y = 155.71, Z = 62.56; box size 25 x 25 x 25 Å; spacing 1.0 Å; exhaustiveness 8. These parameters were applied identically to all 12 docking runs.

5. Molecular Docking

AutoDock Vina (via Tamarind Bio) was used for all docking. Each of the six ligands was docked against both receptors, 12 jobs total, using identical grid box parameters and exhaustiveness 8. Each run produced nine poses; the Mode 1 (most negative) affinity is reported as the best ΔG . Logs were verified to confirm the grid parameters matched inputs.

5.1 Docking Affinity Results

Compound	Role	ΔG vs 1MJE (kcal/mol)	ΔG vs BRCA2_SM (kcal/mol)	$\Delta\Delta G$ (BRCA2_SM – 1MJE)	Rank consistent?
Scytonemin	Test	-10.99	-11.01	-0.02	Yes
Olaparib	Affinity ref*	-9.932	-9.553	+0.38	Yes
Phycocyanobilin	Test	-9.425	-9.081	+0.34	Yes
Lyngbyabellin A	Test	-8.903	-8.272	+0.63	Yes
Nostocarboline	Test	-6.999	-6.327	+0.67	Yes
Caffeine	– Control	-5.444	-5.315	+0.13	Yes

Headline result: Scytonemin (-10.99 / -11.01 kcal/mol) shows stronger predicted affinity than olaparib (-9.932 / -9.553 kcal/mol) on both receptor tracks. All four cyanobacterial compounds maintained identical rank ordering across both tracks, with track-to-track differences of only 0.02-0.67 kcal/mol.

Note: Olaparib is a PARP1/2 inhibitor with no known BRCA2 DBD binding affinity; it was used as a comparative affinity benchmark rather than a validated site binder.

6. Validation

Standard redocking validation (extracting a co-crystallized small-molecule ligand and reproducing its pose) was not feasible here. The co-crystallized partner in 1MJE is DSS1, a 70-residue peptide, which cannot be represented as an SDF or submitted to AutoDock Vina. Three alternative checks were used:

- **Pocket positional check:** For each compound, log files were confirmed against the defined grid box, and Mode 1 pose coordinates were verified to fall within the box.
- **Negative control discrimination:** Caffeine (no known BRCA2 binding) scored -5.444 (Track 1) and -5.315 kcal/mol (Track 2). All four test compounds scored more negative, distinguishing them from a non-binder.
- **Cross-receptor consistency:** Identical rank ordering across both tracks ($\Delta\Delta G$ 0.02-0.67 kcal/mol) indicates results are not artefacts of a single receptor preparation.

Supporting receptor metrics: structural alignment RMSD 0.324 Å; QMEANDisCo Global 0.64 ± 0.05.

7. Binding Pose & Interaction Analysis

7.1 Pose Visualization (PyMOL)

Top-ranked poses were visualized in PyMOL against 1mje_clean.pdb, with the original DSS1 position loaded as a reference to confirm each ligand occupied the correct groove. Contact residues within 4 Å of each ligand were identified and cross-referenced against the PrankWeb P1 residue list. Pocket-overlap ranged from 58% to 92%.

Compound	Contacts (≤4 Å)	P1 pocket overlap
Scytonemin	13	12/13 (92%)
Phycocyanobilin	12	7/12 (58%)
Lyngbyabellin A	10	6/10 (60%)
Nostocarboline	9	8/9 (89%)

Track 2 pose visualization was not performed. The SWISS-MODEL simulation output contains alternate-conformation records that prevent PyMOL from rendering a clean cartoon. Track 2 is therefore reported as affinity values only.

7.2 Interaction Diagrams (PLIP)

PLIP was run on the DOCKED complex files (Track 1) for the four cyanobacterial compounds. TRP2550 and LYS2551 were contacted by all four — the core anchor residues of the P1 pocket.

Compound	ΔG	Key interactions
Scytonemin	-10.99	7 hydrophobic, 1 H-bond (LYS2551), 2 pi-stacking (TRP2550), 3 pi-cation (LYS2551, LYS2671) — richest profile
Phycocyanobilin	-9.425	8 hydrophobic, 1 H-bond (LYS2671; strongest geometry, D-A 2.91 Å)
Lyngbyabellin A	-8.903	5 hydrophobic, 2 halogen bonds via Cl (CYS2610, GLY2669) — only compound with halogen bonds
Nostocarboline	-6.999	3 hydrophobic, 3 pi-stacking (TRP2550) — consistent with beta-carboline aromatic system

8. ADMET Profiling

Drug-likeness and pharmacokinetics were assessed with SwissADME (all six SMILES in one batch); toxicity was predicted with ADMETlab 3.0 (per-compound). ADMETlab outputs probabilities (0-1); values above 0.5 indicate a predicted positive.

8.1 SwissADME — Drug-likeness & Pharmacokinetics

Parameter	Scytonemin	Phycocyanobilin	Lyngbyabellin A	Nostocarboline
MW (g/mol)	544.55	586.68	691.69	217.67
Lipinski violations	1	1	2	0
GI absorption	Low	Low	Low	High
BBB permeant	No	No	No	Yes
Bioavailability score	0.55	0.17	0.17	0.55
Synthetic accessibility	4.65	6.11	7.78	1.63

8.2 ADMETlab 3.0 — Toxicity (probability)

Parameter	Scytonemin	Phycocyanobilin	Lyngbyabellin A	Nostocarboline
AMES mutagenicity	0.891	0.615	0.925	0.456
Hepatotoxicity	0.970	0.974	0.931	0.330
Carcinogenicity	0.876	0.796	0.948	0.634
Genotoxicity	1.000	1.000	1.000	0.981
hERG blocker	0.267	0.016	0.007	0.288
Nephrotoxicity	0.964	1.000	0.999	0.783

Key ADMET findings. Nostocarboline is the most drug-like compound (0 Lipinski violations, high GI absorption, lowest synthetic-accessibility score, and the lowest toxicity probabilities across all endpoints). Scytonemin has an acceptable bioavailability score but high predicted toxicity. All four compounds show high genotoxicity probability (≥ 0.981) — a critical limitation reinforcing that these are early-stage computational hits requiring extensive toxicological validation.

9. Limitations

- Docking is static and treats the receptor as rigid; results are hypothesis-generating, not predictive of true binding affinity.
- The human receptor is a homology model built from the mouse 1MJE template, so cross-track agreement is partly expected by construction — not fully independent validation.
- Track 2 (human model) pose visualization and PLIP interaction analysis were not performed due to alternate-conformation formatting of the SWISS-MODEL output; Track 2 is reported as affinity values only.

- Redocking validation was not feasible (the co-crystallized partner DSS1 is a peptide, not a small molecule).
- All four cyanobacterial compounds show high predicted genotoxicity (≥ 0.981); these are in silico predictions requiring experimental confirmation.
- No molecular dynamics simulation was performed — flagged as future work.

10. Summary

Four cyanobacterial compounds were screened against the BRCA2 DSS1-binding pocket across two receptor structures. Scytonemin emerged as the strongest binder, outperforming olaparib on both tracks, with the richest interaction profile (hydrophobic, hydrogen-bonding, pi-stacking, and pi-cation contacts). Rank ordering was identical across both receptors, supporting the robustness of the affinity results. Nostocarboline, though the weakest binder, presented by far the most favourable ADMET profile. High predicted genotoxicity across all compounds means these results are early-stage computational hypotheses requiring experimental follow-up.

Tools: AutoDock Vina (Eberhardt et al. 2021; Trott & Olson 2010); P2Rank/PrankWeb (Krivak & Hoksza 2018); SWISS-MODEL (Waterhouse et al. 2018); PLIP (Salentin et al. 2015); SwissADME (Daina et al. 2017); ADMETlab 3.0 (Gui et al. 2024); PDB2PQR (Jurrus et al. 2018; Dolinsky et al. 2007); PyMOL (Schrodinger LLC).